

Online supplementary material: ESTA item deletion log

1. Purpose

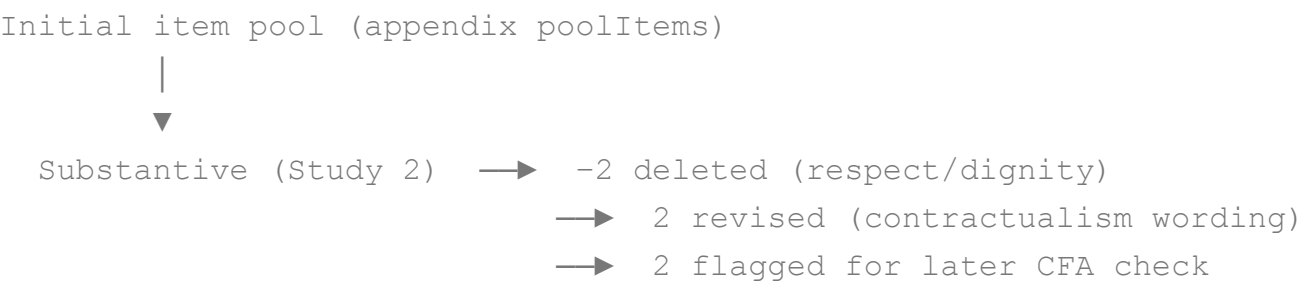
The main text reports only items that were **deleted across all three Study 3 technologies** after confirmatory factor analysis (CFA). Reviewers requested a fuller log: (a) which items were removed or revised at each validation phase, (b) by technology and statistical/judgmental criterion, and (c) what happened to items that showed problems in only one or two technologies.

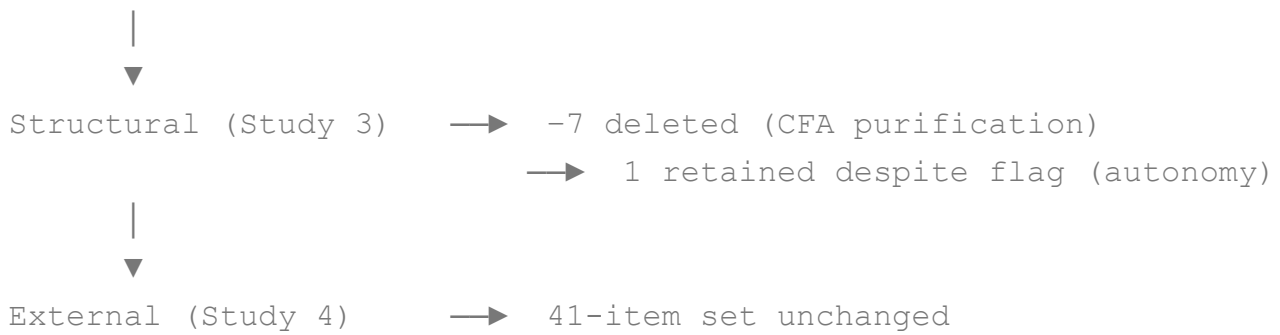
This document provides that trail. **External validity (Study 4) did not delete further items**; the purified 41-item set was used as-is.

2. Overview of the purification pipeline

ESTA validation followed three conventional phases of construct validation. Item content changed only in the first two:

Phase	Study	Technology context	What happened to items
A. Substantive validity	Study 2 (web probing, N = 92)	SAI narrative only	2 items deleted ; 2 revised ; 2 flagged (not yet deleted)
B. Structural validity	Study 3 (CFA, N = 297)	Nano-Pat-Parka (NPP), HomeMate (HM), SAI narrative	7 items deleted when problems were consistent across all three technologies; 1 item retained despite a flag
C. External validity	Study 4 (SEM, N = 579)	SAI descriptive	No item deletions – 41 retained items used in second-order CFA / SEM





3. Decision rules (structural phase)

Item deletion in Study 3 followed an iterative scale-purification process integrating **statistical** and **judgmental** criteria (Wieland et al., 2017, 2018), as described in the manuscript (§3.3):

Criterion type	Indicators used
Statistical	Low explained variance (R^2), high residual variance, large residual correlations / modification indices (local dependence), modification indices / loadings suggesting cross-loadings
Judgmental	Theoretical representativeness, semantic distinctiveness, parsimony, alignment with the ethical-theory definition

Deletion rule used in the manuscript: Only items that **consistently** showed statistical problems **across all three technologies** (NPP, HM, SAI) were removed. Items with concerns in only one or two technologies were retained (see [Section 8](#)).

Flagging thresholds:

Flag	Threshold
Low standardized loading	< .40
Low R^2 (communality)	< .30
Residual correlation / dependence	$ r \geq .10$ (and/or modification index ≥ 10 for residual covariances)
Cross-loading	non-target loading $\geq .30$

4. Item ID lineage (important)

The same numeric suffix can refer to **different stems** across phases. Supplementary tables therefore distinguish **pool IDs** (appendix initial set / Study 2) from **Study 3 column IDs** (data file / CFA).

4.1 Known collision: deontology07 / deontology08

Phase	ID	Stem (short)	Decision
Study 2 (pool)	deontology07	Respect in interactions	Deleted (comprehension)
Study 2 (pool)	deontology08	Dignity in interactions	Deleted (comprehension)
Study 3 (column)	deontology07	Ecological environment (pool deontology10)	Deleted (residual redundancy with nature item)
Study 3 (column)	deontology08	Cultural environment (pool deontology11)	Retained

After probing removed pool respect/dignity items, remaining deontology stems were renumbered in Study 3:

Pool ID (appendix)	Study 3 ID	Stem (short)
deontology09	deontology06	Intrinsic value of nature
deontology10	deontology07	Ecological environment
deontology11	deontology08	Cultural environment

5. Phase A – Substantive validity (Study 2)

Design. Web probing with category-selection and comprehension probes on 21 ESTA items; technology = **SAI narrative** only.

5.1 Items deleted

Pool ID	Theory	Stem (short)	Criterion	Justification (summary)
deontology07	Deontology	Respect in interactions	Comprehension confusion	Participants struggled with “treatment with respect”; answers drifted to consent, non-maleficence, harm avoidance
deontology08	Deontology	Dignity in interactions	Comprehension confusion	Uncertainty about “dignity”; many questioned relevance for SAI (little person-to-person interaction)

5.2 Items revised (kept, reworded)

Pool ID	Criterion	Original focus	Final wording (appendix finalItems)
contractual ist03	Comprehension confusion	Unwritten contract	Supports / works against implicit moral conventions
contractual ist04	Comprehension confusion	Unspoken promise	Violates / does not violate important social norms

5.3 Items flagged but not deleted in Study 2

Pool ID	Criterion	Note	Later outcome (Study 3)
hedonis m02	Ambiguous interpretation	"Selfish" read as personal vs societal scope	Deleted (low R^2)
hedonis m03	Ambiguous interpretation	"Sacrifices" read as personal vs collective	Deleted (low R^2 / residual dependence)

Probe coverage for all probed items: [table_probing_coverage.csv](#) . Author decision list: [table_author_decisions.csv](#) .

6. Phase B – Structural validity (Study 3)

Design. Within-subjects ethical evaluation of three technologies ($N = 297$): Nano-Pat-Parka (**NPP**), HomeMate (**HM**), SAI narrative (**SAI**). Uni-factor CFAs per ethical theory $\times$ technology, then six-factor models; purification as in §3.

6.1 Master structural deletion log (item $\times$ technology $\times$ criterion)

Only the seven items below were **removed**. Columns **R^2 (NPP / HM / SAI)** are from uni-factor CFAs on the pre-purification item set ([table_cfa_item_stats.csv](#)). R^2 rank = rank within that theory's items for that technology (1 = lowest).

Deontology

Study 3 ID	Pool ID (if remapped)	Stem (short)	Criterion	R^2 NPP (rank)	R^2 HM (rank)	R^2 SAI (rank)	Decision
deon tolo gy01	same	Moral obligation to act in a certain way	Low R^2	.395 (1)	.293 (1)	.434 (1)	Deleted

Study 3 ID	Pool ID (if remapped)	Stem (short)	Criterion	R^2 NPP (rank)	R^2 HM (rank)	R^2 SAI (rank)	Decision
deontology07	pool deontology10	Ecological environment	Residual correlation with deontology06 (nature)	.699 (6)	.516 (2)	.772 (8)	Deleted

Residual MI / dependence deontology06 ~~ deontology07 : NPP .34; HM .40; SAI .45.

Judgmental note: nature-focused content redundancy (local dependence).

Hedonism

Study 3 ID	Stem (short)	Criterion	R^2 NPP (rank)	R^2 HM (rank)	R^2 SAI (rank)	Decision
hedonism02	Selfish for me to use	Low R^2 (+ probing ambiguity)	.232 (1)	.443 (2)	.383 (2)	Deleted
hedonism03	Requires sacrifices to use	Low R^2 / residual dependence	.597 (2)	.382 (1)	.290 (1)	Deleted
hedonism09	Harms/promotes my safety	Residual correlation with health/pleasure items; weak conceptual fit to hedonism	.632 (3)	.628 (6)	.797 (8)	Deleted

Example residual flags involving hedonism09 : with hedonism07 (health) MI-implied dependence NPP .72, HM .33, SAI .31; also with pleasure-related items.

Virtue ethics

Study 3 ID	Stem (short)	Criterion	R^2 NPP (rank)	R^2 HM (rank)	R^2 SAI (rank)	Decision
virtue06	Professional excellence	Low R^2 ; within-factor residual correlations	.494 (2)	.375 (1)	.453 (1)	Deleted
virtue07	Respectful towards nature	Cross-loadings to Deontology / Utilitarianism	.493 (1)	.515 (2)	.580 (4)	Deleted

No structural deletions for **relativism**, **contractualism**, or **utilitarianism**.

6.2 Compact decision summary (structural)

Study 3 ID	Technologies affected	Primary criterion	Final action
deontology01	NPP, HM, SAI	Low R^2	Deleted
deontology07 (= pool ecological)	NPP, HM, SAI	Residual dependence	Deleted
hedonism02	NPP, HM, SAI	Low R^2	Deleted
hedonism03	NPP, HM, SAI	Low R^2 / residuals	Deleted
hedonism09	NPP, HM, SAI	Residual dependence + conceptual	Deleted
virtue06	NPP, HM, SAI	Low R^2	Deleted
virtue07	NPP, HM, SAI	Cross-loading	Deleted

7. Retained despite a statistical flag

Study 3 ID	Stem (short)	Flag observed	Why retained
deontology02	Promotes autonomy of users	Indications of cross-loading with Hedonism; residual flags in 2/3 technologies	Autonomy is central to deontological ethics; item kept on theoretical grounds

8. Net inventory

Stage	Count (approximate)	Notes
Initial theoretical pool	51 items (appendix poolItems)	Relativism 5, contractualism 6, hedonism 9, utilitarianism 10, deontology 11, virtue 9
After substantive deletions	-2	Pool deontology07 , deontology08 (respect/dignity)
Fielded in Study 3	46 items × 3 technologies	e.g. utilitarian10 not in Study 3 data; deontology renumbered after probing
After structural deletions	-7	See §6.2

Stage	Count (approximate)	Notes
Final validated set	41 items	Mplus <code>secondOrder_ESTA.inp</code> <code>USEVARIABLES ;</code> appendix finalItems (some stems renumbered for publication)
Study 4 external validity	41 items	No further deletions